

**Root Duality: Unifying Indexical Identity and Computational Anchoring —  
A Focus on the Mechanisms of Complex Word Formation**

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**Abstract:**

This paper analyzes the study of root properties in constructional morphology, examining the multilingual process of complex word construction. Building on Belder & Craenenbroeck (2015), which characterizes the essence of roots as the “empty position” created by initial Merge-driven displacement, the paper presents a systematic analysis and argues that roots exhibit a dual nature as both indexical units and computational anchors.

**Key words:** root, empty position, index, Computational Anchoring

## **1. Introduction**

### **1.1 Lexicalism and Neo-Constructivism in the Study of Word Formation**

Within generative grammar, research on word formation has long been organised around the question of whether there exists a lexicon, or, more precisely, how the relation between the lexicon and syntax ought to be theorised. Two broad orientations have consequently emerged: Lexicalism and Neo-Constructivism. Since Chomsky’s (1970) formulation of the Lexicalist Hypothesis, lexicalist approaches have tended to maintain a principled division between the lexicon and syntax. On this view, word-formation rules—derivation, compounding, and related morphological operations—are completed within the lexicon as an autonomous computational component before syntactic derivation begins. The lexicon is thus endowed with both generative capacity and relative autonomy: syntax may operate only on already-formed words, while the internal constitution of such words remains inaccessible to syntactic computation, as captured by the Lexical Integrity Hypothesis.

By contrast, Neo-Constructivist approaches—most notably those associated with Distributed Morphology and related work (Halle & Marantz 1993, 1994; Embick

2004; Borer 2013)—advance the thesis often glossed as “syntax all the way down”. Under this conception, word formation and syntax are not generated by distinct engines. Rather, both are taken to arise from the same underlying combinatorial mechanism, namely Merge. It is precisely this theoretical move that displaces the lexicalist lexicon from its privileged generative position: what had once been treated as a unified lexical component is decomposed into three independent lists, distributed across different stages from syntax to the interfaces. With the lexicon thus deprived of its original status as the initial site of generation, a new and rather fundamental question comes into view: what, exactly, is the nature of the Root, the putative starting point of syntactic derivation?

## 1.2 Scholarly Debates on the Nature of the Root

Within the existing Neo-Constructivist framework, the ontological status of the Root remains unsettled. What one finds is not yet a systematic consensus, but rather a field marked by sustained theoretical division. In broad terms, three positions may be distinguished: the indexical view, represented chiefly by Harley (2014); the ontological view, associated with Marantz (1997) and Embick (2004); and the structural view, developed in different ways by Acedo-Matellán and Mateu (2015), Borer (2013), and De Belder and Van Craenenbroeck (2015).

Pan Haihua and Ye Kuang (2025) also offer a careful synthesis of the debate on the nature of roots, though their classification differs slightly from the one adopted here<sup>1</sup>. They observe that, when the relevant proposals are considered together, roots appear to possess phonological features, conceptual meaning<sup>2</sup>, ontological meanings such as eventivity and stativity<sup>3</sup>, and even the ability to take complements. Such a profile, they note, comes remarkably close to Lexicalism<sup>4</sup>. Yet, if one considers any single scholar’s position in isolation, the divergences are considerable. Agreement extends only to the claim that roots lack lexical category; on almost every other question, scholars remain divided.

Their assessment, I would suggest, brings into sharper relief a paradox at the centre of the ontological discussion of roots. At the syntactic level, the Root is

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1 I discuss these three views systematically in Chapter 2.

2 I take this position to belong to the ontological view of the nature of roots. My interpretation of the indexical view differs from that of Pan Haihua and Ye Kuang (2025).

3 I take this position, too, to fall under the ontological view of the nature of roots.

4 I take this position to represent the structural view of the nature of roots.

treated as an “empty” object, lacking both syntactic category and grammatical features. At the interfaces, however, it becomes a “substantial” object, subject to semantic interpretation and phonological insertion. It is this tension between syntactic emptiness and interface substantiality that, at least in part, explains why the nature of the Root has remained so persistently disputed.

### 1.3 The Challenge Posed by Chinese Data in Pan Haihua and Ye Kuang (2025)

By comparing strongly morphological languages, represented by English, with weakly morphological languages, represented by Chinese, Pan and Ye (2025) argue that the paradox discussed in Section 1.2 may be handled through morphological operations in the former type of language, whereas in the latter such operations are not readily available. More pointedly, they suggest that four challenges of applicability arise.

(a) **The difficulty of distinguishing roots from words.** In strongly morphological languages such as English, roots are often distinguished from categorised stems through suppletion or derivational morphology, as in *destroy* versus *destruction*. In Chinese, by contrast, many monosyllabic morphemes exhibit no overt morphological distinction between word and root.

(b) **The increase in derivational cost.** From a lexicalist perspective, many simple Chinese words, such as *you-yong* ‘swim’, may be generated directly in the lexicon before entering syntactic computation. Neo-Constructivism, by contrast, must first decompose such items into a set of considerably more intricate syntactic operations. What is otherwise a relatively simple derivational process is thereby rendered complex, with little apparent gain in either economy or explanatory plausibility.

(c) **The indeterminacy of feature valuation.** In Chinese, semantic interpretation often depends on the level at which an expression is treated as a word. Pan and Ye argue that, if one follows a Distributed Morphology-style derivational model, whether expressions such as *jie-ke* ‘receive guests’ and *jie-ke-ren* ‘receive guests/clients’ count as words or phrases is effectively left to encyclopaedic knowledge. The problem, on this view, is not so much resolved as displaced.

(d) **The difficulty of determining primary and secondary functions.** Chinese morphemes are functionally flexible, displaying both typical and non-typical distributions. Pan and Ye suggest that, within a Distributed Morphology

account, it remains difficult to specify which functions are primary and which are peripheral, or to explain why such a hierarchy should obtain in the first place.

The criticism directed at Neo-Constructivist root theory, especially in its Distributed Morphology variant, may thus be understood as follows. If roots or morphemes are inserted directly into syntax in a weakly morphological language such as Chinese, the derivation appears to require an ever-expanding sequence of generative steps, while a substantial portion of the explanatory burden is shifted to the interfaces. Compared with the lexicalist alternative, where words are formed in the lexicon before being handed over to syntax, the Neo-Constructivist procedure can seem less economical and less straightforward.

I would argue, however, that Pan and Ye's four challenges need not entail a wholesale rejection of Distributed Morphology. As their own concluding remarks suggest, their critique in fact exposes two blind spots in Neo-Constructivist root theory.

(a) In the derivation, the Root is often tacitly assumed to be identical to the language state in which it is externalised, while the role of the three lists is consequently underplayed.

(b) Whether the Root is regarded as a featureless index, as an entity endowed with pre-specified ontological semantics, or as a structurally defined syntactic object, such accounts tend to overlook how derivational timing and the division of labour among the interfaces dynamically shape the nature of the Root during syntactic building.

On this basis, the apparently comprehensive challenge posed by the Chinese data in Pan and Ye (2025) to Neo-Constructivist root theory should perhaps be seen in a different light. Rather than closing down the Distributed Morphology programme, it opens a valuable opportunity to examine it with greater precision and to reconsider the Root as a dynamic object. The following sections therefore take these two blind spots as their point of departure, and seek to provide a more fine-grained account of the nature of roots within a Neo-Constructivist framework, with Distributed Morphology serving as the principal point of reference.

## **1.4 Research Aims and Contributions**

Against this background, I suggest that divergent understandings of the Root in the linguistic literature do not necessarily reflect mutually exclusive theories, but

rather different points of theoretical emphasis. Each, in its own way, offers a useful vantage point for examining the nature of roots. The indexical view foregrounds the Root's role in serving interface operations; the ontological view treats the Root as a morphological benchmark; and the meaning-free view, by contrast, emphasises the Root as a semantic benchmark. Taken together, these positions may be understood as accounts of the Root at different stages of syntactic construction. Put differently, as already suggested in Section 1.3, the Root is not a static object with invariant properties throughout language construction, but a dynamic object whose status shifts as derivation proceeds.

This paper accordingly proposes the Duality of Root Hypothesis.

### **(1) The Duality of Root Hypothesis**

The Duality of Root Hypothesis holds that the function of the Root in language generation is dynamic. Different functions are assigned to it at different stages of syntactic construction. More specifically, once syntactic operations have been completed and the derivation is transferred to the Sensory-Motor Interface (S-M) and the Conceptual-Intentional Interface (C-I), the Root assumes a dual role in the process of interface identification: it functions both as an indexical unit and as a computational benchmark.

The remainder of the article is organised as follows. Section 2 proposes a derivational model of Root duality. Section 3 examines the role played by the Root as both indexical unit and computational benchmark in the formation of Chinese compounds, before returning, from a comparative perspective, to the significance of root duality in compounding. Section 4 briefly considers what implications Root duality may have for the study of syntactic building mechanisms. Section 5 concludes the article.

## **2. Theoretical Reconstruction: A Derivational Model of the Duality of Root**

### **2.1 Reconsidering Three Views of the Root**

As noted in Chapter 1, Neo-Constructivist approaches to the ontology of the Root may be broadly divided into three orientations: the indexical view, the ontological view, and the structural view.

The first is the indexical view. Scholars such as Harley (2014) take roots to be indices. In Harley's syntactic representation, the Root is expressed as an arbitrary number or index, whose function is to distinguish each root from all others. A single root, on this account, may receive different phonological realisations in different syntactic environments. Since the semantic content of roots is highly variable, and may even resemble a form of "semantic substitution", neither phonological nor semantic properties can serve as reliable criteria for identifying a root at the initial stage of derivation. Harley therefore proposes that, in syntax, the Root should be treated as a purely structural unit deployed for syntactic computation, without either semantic or phonological features.

The second is the ontological view. Marantz (1997) and Embick (2004), among others, take roots to possess built-in ontological properties. Marantz, for instance, distinguishes entity roots, such as those denoting objects, from state roots, such as those denoting properties, and event roots, such as those denoting actions. Rappaport Hovav and Levin (2010), by a related logic, propose a distinction between manner and result roots, with the semantic type of a root helping to determine the syntactic behaviour of the verb in which it occurs.

The third is the structural view. Acedo-Matellán and Mateu (2015), Borer (2013), and related work argue that the semantic properties of roots are derived from syntactic structure rather than from any intrinsic semantics. On this view, roots are category-free abstract units, and their interpretation depends entirely on the position in which they combine with functional morphemes, such as *v* and *n*.

De Belder and Van Craenenbroeck (2015), proceeding from the principles of Merge, identify four properties of roots: roots have no grammatical features; they have no syntactic category; they are defined structurally rather than lexically; and they are dominated by functional categories, rather than themselves dominating such categories.

These three views may now be assessed more closely. With respect to the indexical view, Harley (2014) treats the Root in syntax as an arbitrarily numbered index, such as #53 or #67. The same root may surface with different phonological forms in different syntactic environments. Given the considerable variability of root meaning, and given its possible resemblance to "semantic substitution", roots cannot, at the initial stage, be identified simply by their phonological or semantic features. It follows that the Root, for Harley, is best understood as a purely

structural unit for syntactic computation, containing neither semantic nor phonological features.

Marantz, however, suggests that Harley's proposal faces pressure from at least two directions.

First, genuinely observable cases of root suppletion are exceedingly limited across languages. This fact sits rather uneasily with the theoretical expectation that suppletion involving roots and suppletion involving functional categories should be handled under a unified mechanism.

Second, recent work in psycho- and neurolinguistics indicates that speakers acquire sound–meaning associations involving phonemic properties of roots, rather than merely those associated with affixes. Such associations resemble the semantic relation found in English *gl-* forms such as *glimmer* and *glitter*, although speakers are not consciously aware of the knowledge involved. Were the Root in syntax no more than an abstract number wholly detached from phonology, these facts would be difficult to explain in a fully convincing manner.<sup>5</sup>

Taken together, Marantz's objection suggests that root indices are not, strictly speaking, the very elements manipulated by syntax. Syntax does not operate over #53, #67, and the like. Rather, the Root should be conceived as an index that is either isomorphic with, or tightly linked to, an underlying phonological form. On this account, the morphological adjustment from *go* to *wen(t)* begins from the index itself. It is for this reason that Marantz takes roots to contain underlying phonological form, and this position constitutes one of the principal motivations for the ontological view.

From the perspective of Chinese, Marantz's revision of the indexical view is particularly instructive. Let us consider the three examples in (2).

|                         |                 |                      |
|-------------------------|-----------------|----------------------|
| (2) a. 一把锁 (yī bǎ suǒ)  | b. 锁门 (suǒ mén) | c. 连锁 (lián suǒ)     |
| 一        把        锁     | 锁    门          | 连        锁           |
| yī        bǎ        suǒ | suǒ   mén       | lián        suǒ      |
| one.NUM ba.CLF lock.N   | lock.V door.N   | chain.connect-lock.N |

The examples in (2) show that the Chinese form *suǒ* 'lock' may occur in nominal position, as in (2a), in verbal position, as in (2b), and may also participate in compound formation, as in (2c). Within a lexicalist framework, such items would

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<sup>5</sup> This view is based on my email correspondence with Marantz.

typically be pre-encoded in the lexicon with multiple categorial specifications, and would consequently be stored as a set of homophonous lexical entries. Under the indexical view, by contrast, they would be treated as several wholly arbitrary numbers. Yet this apparent simplification comes at a price: syntax does not, in any substantive sense, recognise these arbitrary indices. What syntax recognises is the categorising morpheme with which the relevant element merges. As a result, it becomes rather difficult to establish any necessary connection with List 2.

Marantz's revision<sup>6</sup> is illuminating precisely because it reconnects, to some extent, the index with the Root's basic phonological form and semantic potential. By assuming that the Root may serve, in List 2 and List 3, as a retrieval index for phonological form and encyclopaedic information, the root index in syntax is no longer a wholly arbitrary number. Rather, it bears a traceable correspondence to particular patterns of sound and meaning. As Marantz stresses, language users often establish certain sound–meaning associations at an unconscious level. This provides empirical support for sound–meaning correlations at the level of the Root. Yet it also raises a further question: is it necessary to presuppose, already at the syntactic stage, that the Root bears some form of “basic sound–meaning” content? Put differently, might the phonological and semantic properties of the Root be strictly confined to the interfaces, being gradually realised through the operations of List 2 and List 3, rather than being treated in narrow syntax as an already specified “matrix of sound–meaning features”?

With respect to the meaning-free view, Borer (2013), in the Exo-Skeletal model, argues that roots have no meaning whatsoever. Root meaning is instead obtained, at a certain stage of syntactic derivation, through a search of encyclopaedic information. This position is consonant with the four properties of roots proposed by De Belder and Van Craenenbroeck (2015).

De Belder and Van Craenenbroeck (2015) argue, within the Minimalist Program and Distributed Morphology, that roots are defined by syntactic structure rather than by the lexicon. Their proposal of Asymmetric Primary Merge holds that Primary Merge is directional in the process of syntactic building. More specifically, a set of grammatical features, such as  $\{\alpha\}$ , always merges with the empty set  $\emptyset$ , yielding the ordered pair  $\langle \{\alpha\}, \emptyset \rangle$ , which is then labelled by the projection of  $\{\alpha\}$ . The empty position generated by Primary Merge is the site into which the Root is

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<sup>6</sup> This point is likewise based on my email correspondence with Marantz.



inserted at a later stage<sup>7</sup>. On this account, the Root is a structurally defined “empty position”, bearing neither intrinsic grammatical information nor category, and receiving its category only through the operation of a categorising morpheme.

They further discuss and seek to establish four properties of roots: (a) roots have no grammatical features; (b) roots have no syntactic category; (c) roots are defined by structure, rather than by the lexicon; and (d) roots are dominated by functional categories, rather than the reverse.

Taken together, the indexical view and the ontological view appear to answer different questions. The former concerns the function of the Root at Spell-Out, whereas the latter concerns the content accessed by the root index at the interfaces, with particular emphasis on the role of the Vocabulary List. The meaning-free view, in turn, brings into sharper focus the function of encyclopaedic knowledge. De Belder and Van Craenenbroeck’s (2015) account of Null Derivation, meanwhile, appears to explain how the Root is manipulated during syntactic building.

## **2.2 A Revised Duality of Root Hypothesis**

As the preceding discussion suggests, the differences among the indexical, ontological, and structural views do not necessarily arise from irreconcilable theoretical commitments. They may instead reflect different emphases in the study of roots. Each offers a non-trivial point of reference for clarifying the nature of the Root. The indexical view foregrounds the Root’s function in serving interface operations; the ontological view treats the Root as a morphological benchmark; and the meaning-free view emphasises the Root as a semantic benchmark. Existing discussions, including the critiques of these positions, have nevertheless tended to proceed from a static perspective, assuming that the Root possesses a single fixed property throughout the entire derivation and that this property bears directly on the final outcome of syntactic building. Under such an assumption, one is forced to choose among the three views in an exclusionary manner, thereby giving rise to what Pan Haihua and Ye Kuang (2025) describe as the “agnosticism of root identity”.

Once derivational timing and the division of labour among the interfaces are introduced into the discussion, a different possibility emerges. The Root may be a structural empty position during syntactic building, an index at Spell-Out, and a morphological or semantic computational benchmark at the interfaces. Put differently, the indexical, ontological, and structural views need not be mutually

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<sup>7</sup> On the late insertion of roots, see Harley (2014: 244).

exclusive. They may instead be understood as different functional projections of the same object at distinct stages of generation. It is precisely here that the theoretical point of departure for the Duality of Root Hypothesis lies.

On the basis of the foregoing discussion, the Duality of Root Hypothesis proposed in Section 1.4 may now be revised as follows.

### **(3) The Revised Duality of Root Hypothesis**

At the stage of syntactic building, where List 1 supplies the syntactic terminals for syntactic operations, the Root occupies no more than an empty position introduced by Primary Merge. It contains neither phonological nor semantic content. Once operations such as Merge and Agree have applied, this empty position is converted, at Spell-Out, into an indexical unit. At the stage of morphological operation, where List 2 supplies the Vocabulary Items, ordered linearisation is then carried out. Here, the Root functions as the anchor for morphological alternation. More specifically, the Root, associated after Spell-Out with a basic phonological form, serves as the point of reference at the morphological and phonological interfaces, from which subsequent alternations proceed<sup>8</sup>. Similarly, at the encyclopaedic stage, where List 3 supplies encyclopaedic knowledge, the Root functions as the operational anchor for semantic computation. This model is represented in Figure 1.

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<sup>8</sup> That is, the morphological adjustment from *go* to *wen(t)*, as Marantz notes, begins from the “indices”. The Root therefore has an underlying phonological form, although the insertion of this phonological form is, of course, also delayed.

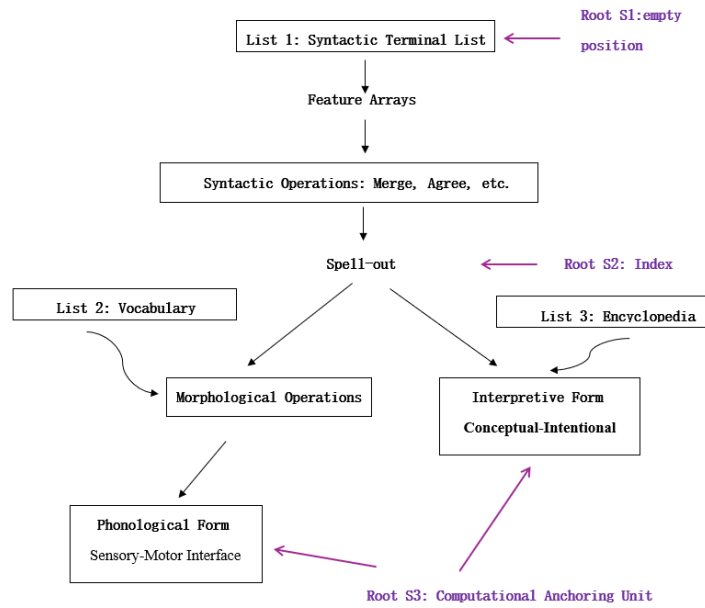


Figure 1 schematically represents this phase-based derivational process.

### 2.3 The First Dimension of Root Duality: Structural Empty Position and Index

The revised Duality of Root Hypothesis treats the stage of syntactic building—where List 1 supplies the syntactic terminals for syntactic operations—in terms of the theory of Null Derivation proposed by De Belder and Van Craenenbroeck (2015). On this view, what emerges at the stage of Primary Merge is a form of Null Derivation. That is, at the initial moment of syntactic building, no substantive lexical item has yet been introduced into the derivation. Structure begins with merger with the empty set; consequently, the derivation is, at its earliest stage, “empty”, bearing neither phonological nor semantic content.

Following Marantz (2007), we may also treat the word as a phase. The feature set of the categorising morpheme functions as the head of this phase, while the material externally merged with it—thereby forming the Spell-Out domain—is transferred to the interfaces. Building on Marantz (2007) and De Belder and Van Craenenbroeck (2015), I further propose that, under Primary Merge, the grammatical feature set involved in Null Derivation, such as  $\{\alpha\}$ , and the empty set  $\emptyset$  together constitute a phase. The head of this phase is the grammatical feature set that undergoes Primary Merge with the empty set, while the component transferred to the interfaces is the  $\emptyset$  node. It follows that the empty position generated at the initial stage of narrow syntactic building is, in essence, an empty set. Once the

derivation reaches Spell-Out, this empty position forms an index and is transferred to the interfaces.

The empty position generated at this initial stage—which I term the Syntactic Position Structural Gap of the Root ( $\text{ROOT}_{\text{SPG}}$ )—therefore has the following properties.

(a) *Categorial absence.* This property is relatively uncontroversial in the literature. According to Null Derivation theory, the Root is a structural empty position generated through initial syntactic merger. Only after merging with a categorising morpheme does it acquire a grammatical category. It follows that this structural empty position may merge with any functional head.

(b) *Semantic absence.* As a structural empty position, the Root naturally carries no semantic information. At Spell-Out, what it forms is only an “empty index”: a structural address marked as requiring semantic filling at a later stage, rather than a concrete concept or event type.

(c) *Licensing of structures without functional heads.* Since the Root is a syntactic structural empty position generated by initial merger, the direct merger of two roots does not yield an unlabellable configuration of the form  $\{\text{XP}, \text{YP}\}$ . Roots carry no features; consequently, the absence of a functional head is licensed at the level of Merge itself. The structure so licensed is, in effect, a larger indexed empty set. From a set-theoretic perspective, no matter how many roots are merged, only one set is generated at the stage of syntactic building.

At this point, the Root may be divided into two stages. The first stage (S1) is the structural empty position of syntactic building, namely  $\text{ROOT}_{\text{SPG}}$ . The second stage (S2) is the Index Unit, namely  $\text{ROOT}_{\text{IU}}$ . The index at S2 is, strictly speaking, only a “position”. After Primary Merge, the component transferred to the interfaces is an index slot created within the workspace. Here I take the phase containing the Root to constitute a workspace, whose head is a categorising morpheme. The function of this index slot<sup>9</sup> is to indicate to the interfaces that there is a position

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<sup>9</sup> Halle (1992) construes Spell-Out as the rewriting of a place-holder “Q” in a morpheme as phonological material. Halle (2014) later treats this place-holder as an index. Professor Cheng Gong has pointed out, in relation to this issue, that the dynamic character of the Root may be problematic, since syntactic operations should involve atomic elements. My response is as follows. Following Marantz (2007), I take the feature set merged with the Root, when transferred to the interfaces, to write its own features into the empty set; in other words, it transmits phasehood to that empty set. The empty set thereby yields an index at Spell-Out

requiring later filling. This conception is rather close to the box theory adopted by Chomsky (2024). Functionally, the index slot may likewise be understood as a kind of box. The functional head merged with it can access this index <sup>10</sup>and write features into it within the workspace, or phase. Subsequent interface processing may then proceed through regular constituent-building.

A further question then arises: how are affixed words formed under this model? With respect to affixes, Creemers, Don, and Fenger (2018) propose that some affixes are roots, whereas others are categorising morphemes, a claim they support with evidence from Dutch. Since compounding constitutes one of the major word-formation strategies in Chinese (Chen & Liu 2020), Chapter 3 will address this issue through Chinese compound data.

## **2.4 The Second Dimension of Root Duality: Computational Anchoring**

In Section 2.3, drawing on the notion of derivational timing, I characterised the Root at the first and second stages as, respectively, a structural empty position (ROOT<sub>SPG</sub>) and an index unit (ROOT<sub>IU</sub>). On that basis, the present section turns to the third stage, at which the Root functions as a computational anchor, or ROOT<sub>CA</sub>.

The third stage begins after syntactic building has been completed, when the entire Spell-Out domain undergoes Transfer. At the interfaces, the Index Unit (ROOT<sub>IU</sub>) now functions as a computational anchor for phonological insertion, and, at the encyclopaedic interface, as a computational anchor for conceptual and semantic interpretation. The former concerns an underlying phonological representation supplied through the Vocabulary List. This stage provides the starting point for language-specific morphological rules. The alternation from *go* to *went*, for instance, may be understood as the externalised outcome of underspecification and competition for insertion at the morphological interface, operating over features carried by the Index Unit (ROOT<sub>IU</sub>).

With regard to encyclopaedic knowledge, Harley (2012) observes that, at the LF interface, information is accessed in addition to ordinary mechanisms of

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and is subsequently valued at the interfaces. On this analysis, there is no direct conflict with the requirement that syntactic operations involve atomic elements.

<sup>10</sup> I take Chomsky's (2024) box theory to be a further development of the transmission of phasehood proposed in Chomsky (2013, 2015).

semantic composition in order to interpret roots in specific syntactic-semantic environments. This information is stored in the third list, namely the Encyclopedia or List 3. Both idiomatic and conventional meanings of roots are stored there, and encyclopaedic information works together with the semantic denotations of *f*-morphemes to generate the final interpretation of the structure.

This, however, is the more traditional formulation. More recent work in Distributed Morphology has often entertained the possibility that functional morphemes, too, may display a root-like form of contextual allosemy (Marantz 2013; Harley 2014; Embick 2016). Put differently, the real diagnostic for encyclopaedic knowledge in Distributed Morphology is not “roothood” as such, but unpredictability. Any information that cannot be derived from syntactic features—what the Minimalist Program would place under the third factor—must be supplied by encyclopaedic knowledge. Embick (2016), for instance, suggests that encyclopaedic knowledge not only supplies conceptual content for roots, but may also provide non-compositional interpretations for particular functional features, such as gender and number, in specific contexts.

We may therefore conclude that, at the Encyclopedia, the Root activates access to encyclopaedic knowledge. It defines the entry point for semantic interpretation. Since semantic filling remains subject to locality constraints, and since the relevant interpretation is fixed only after semantic negotiation with adjacent functional nodes, the Root at this stage is best understood as the computational anchor for semantic filling.

## 2.5 Summary

This chapter has revised the Duality of Root Hypothesis by introducing derivational timing and the division of labour among the interfaces into the dynamic construction of root ontology. In doing so, it departs both from the holistic treatment of word formation in traditional lexicalism and from the Neo-Constructivist assumption that roots are pre-fabricated static operators. The Root is instead defined as a polymorphic operator that evolves dynamically as derivation proceeds. The process of this dynamic heterogenisation may be summarised as follows.

At the stage of syntactic building, the Root is a structural empty position (ROOT<sub>SPG</sub>). It carries no determinate phonological or semantic information, and is formalised only as a “vacant” ordered-pair existence:

$$ROOT_{SPG} \in \kappa: \kappa = \langle \emptyset_{\pi}, \emptyset_{\sigma} \rangle_{\text{Structure}}$$

When syntactic building is completed and the derivation is transferred to Spell-Out, the structural empty position ( $ROOT_{SPG}$ ) is dynamically converted into an Index Unit ( $ROOT_{IU}$ ). In other words,  $ROOT_{SPG}$  is transformed into an indexical address recognisable at the interfaces:

$$\mathcal{F}_{idx}: ROOT_{SPG} \xrightarrow{\Phi_{\text{Spell-out}}} ROOT_{IU} \in \Pi$$

The Root is then further converted into a computational anchor for morphological, phonological, semantic, and encyclopaedic computation ( $ROOT_{CA}$ ) :

$$ROOT_{CA} = \begin{cases} \Phi_{\text{Form}} \models \langle i, C \rangle \\ \Sigma_{\text{Sem}} \models \langle i, W \rangle \end{cases}$$

In this formalisation, (i) is the unique index generated at Spell-Out. The function ( $\mathcal{F}_{idx}$ ) converts the structural empty position ( $ROOT_{SPG}$ ) into an Index Unit ( $ROOT_{IU}$ ) recognisable at the interfaces. (C), or Contextual Constraint, denotes the set of local morphophonological environmental features. (W), or World Knowledge/Encyclopedia, denotes the encyclopaedic knowledge base. ( $\Phi_{\text{Form}}$ ) is the formal-representational anchoring operator, while ( $\Sigma_{\text{Sem}}$ ) is the semantic-interpretive anchoring operator. The former requires the phonological representation corresponding to index (i) to be realised within a local morphophonological feature environment; the latter indicates that semantic interpretation is constrained by world knowledge, thereby completing the mapping from an abstract index to a concrete conceptual core.

The derivational chain of the Root's dynamic evolution under the revised Duality of Root Hypothesis may therefore be summarised as follows.

#### (4) The Derivational Chain of the Root's Dynamic Evolution

$$ROOT_{SPG} \xrightarrow{\text{Spell-out}} ROOT_{IU} = i \xrightarrow{\text{Transfer}} ROOT_{CA}\{\Phi(i, C)\Sigma(i, W)\}$$

## 3. Re-examining the Duality of Root Hypothesis

### 3.1 Why Chinese Compounds?

Chinese compounds are selected here as the crucial empirical domain for testing the Duality of Root Hypothesis not because Chinese is typologically exceptional in any simple sense, but because a language of this kind reduces certain forms of interference that might otherwise obscure the analysis. Chinese lacks overt categorising affixes and case markers. This property strips away, to a considerable extent, morpho-phonological interference, thereby providing an almost ideal empirical environment in which the structural behaviour of roots may be examined.

The cross-categorical phenomena found in Chinese, such as those illustrated in (2), are particularly revealing in this respect. Any difference in morphological function cannot be attributed to surface morphological marking, but must instead be traced to differences in the derivational path. In this way, the mergeability of Chinese roots at the level of syntactic building can be examined separately from their atomicity at the interfaces. It is precisely this separability that provides salient evidence for the dual properties of the Root.

At the same time, Chinese compounds expose with some clarity the explanatory limitations of certain strands within the Neo-Constructivist framework. They also offer a useful point of entry for responding to recent criticisms, especially those advanced from a lexicalist perspective. As Pan Haihua and Ye Kuang (2025) argue, traditional work in Distributed Morphology has not always been able to distinguish effectively between root and word, or between word and phrase. Existing studies have generally taken one of two routes: either compounds are reduced entirely to syntactically manipulated phrases, as in Zhang, N. N. (2007), or their construction is assigned to the interfaces.

This article argues that the challenge raised by Pan and Ye (2025) in fact reveals the necessity of introducing a dynamic approach to roots. The study of Chinese compounds compels us to move away from a static conception of root ontology and towards a derivational perspective shaped by derivational timing and the division of labour among the interfaces.

### **3.2 Lexicalisation Tests for Four Types of Compound Structure**

### **3.3 The Derivation of Chinese Compound Forms under the Duality of Root Hypothesis**

### **3.4 The Nature of Compound Roots from a Comparative Perspective**

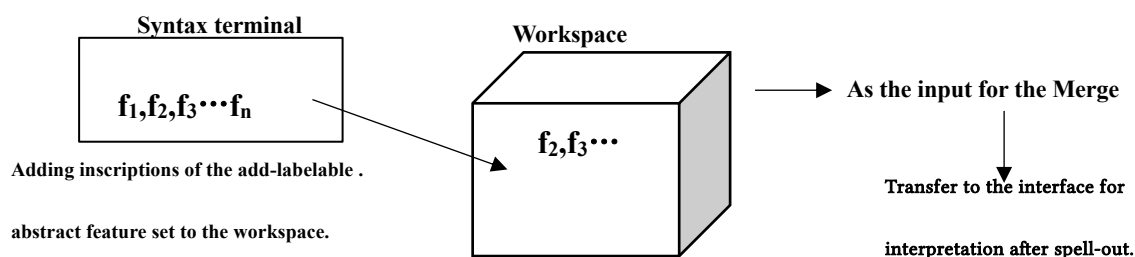


### 3.5 Summary

## 4. Implications of Root Duality for Syntactic Building in Neo-Constructivism

The investigation of root duality suggests that the method involved in word-building is, in effect, also a form of phase-based derivation within syntactic building. What must therefore be reconsidered is the nature of the objects on which syntactic building operates. As Heidi Harley (2021) observes, the primitive elements of syntactic building are abstract feature sets of subtypes. I therefore take syntactic building to operate precisely over such abstract subtype feature sets.

When word-building is projected onto the dimension of syntactic building, what is in fact sent into the workspace is the inscription of labellable abstract feature sets. These inscriptions undergo Merge, or Agree, within the workspace, and are subsequently transferred to the interfaces for interpretation, as shown in Figure 2.



**Figure 2. Language merging construction mechanism under Neo-constructionism**

The foregoing analysis has deliberately adopted the term “building” rather than the more familiar term “generation”. The reason is not merely terminological. Human beings are not born with knowledge of any particular language—that is, of any specific native language—although they may be endowed with a rudimentary language organ. The gradual maturation of this organ, through which particular languages come to be acquired, is what may more properly be called “generation”.

By contrast, when we analyse word formation and sentence formation, what is at issue is a different process. Once the elements of the syntactic terminal list have been inscribed, they are built into structures, which are then handed over to the Vocabulary List and the Encyclopedia. These two lists complete the “optimal

selection” involved in underspecification, before the resulting structure is finally interpreted. For this reason, I use “building” to refer to the syntactic and lexical operations involved in word formation, rather than “generation”.

It should also be made explicit that the Root is not, strictly speaking, present on the syntactic terminal list<sup>11</sup>. Rather, it is the empty position generated through Merge: an index is assigned to it at Spell-Out, and at the interfaces it functions as the anchor for building computation.

## 5. Conclusion

Through an examination of the nature of the Root, this article has proposed that the Root performs different functions at different stages of building. It has also advanced the view that the syntactic terminal list in Distributed Morphology should not include the Root. This, I suggest, constitutes a non-trivial revision of the standard picture.

The article has also distinguished “generation” from “building”. On this view, when word formation and sentence formation are under discussion, “building” is the more appropriate term. It captures the constructional operations by which inscribed abstract feature sets are assembled, transferred, selected, and interpreted, whereas “generation” is better reserved for the maturational process through which the human language faculty comes to support the acquisition of particular languages.

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<sup>11</sup> That is, the syntactic terminal list contains only abstract morphemes.

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